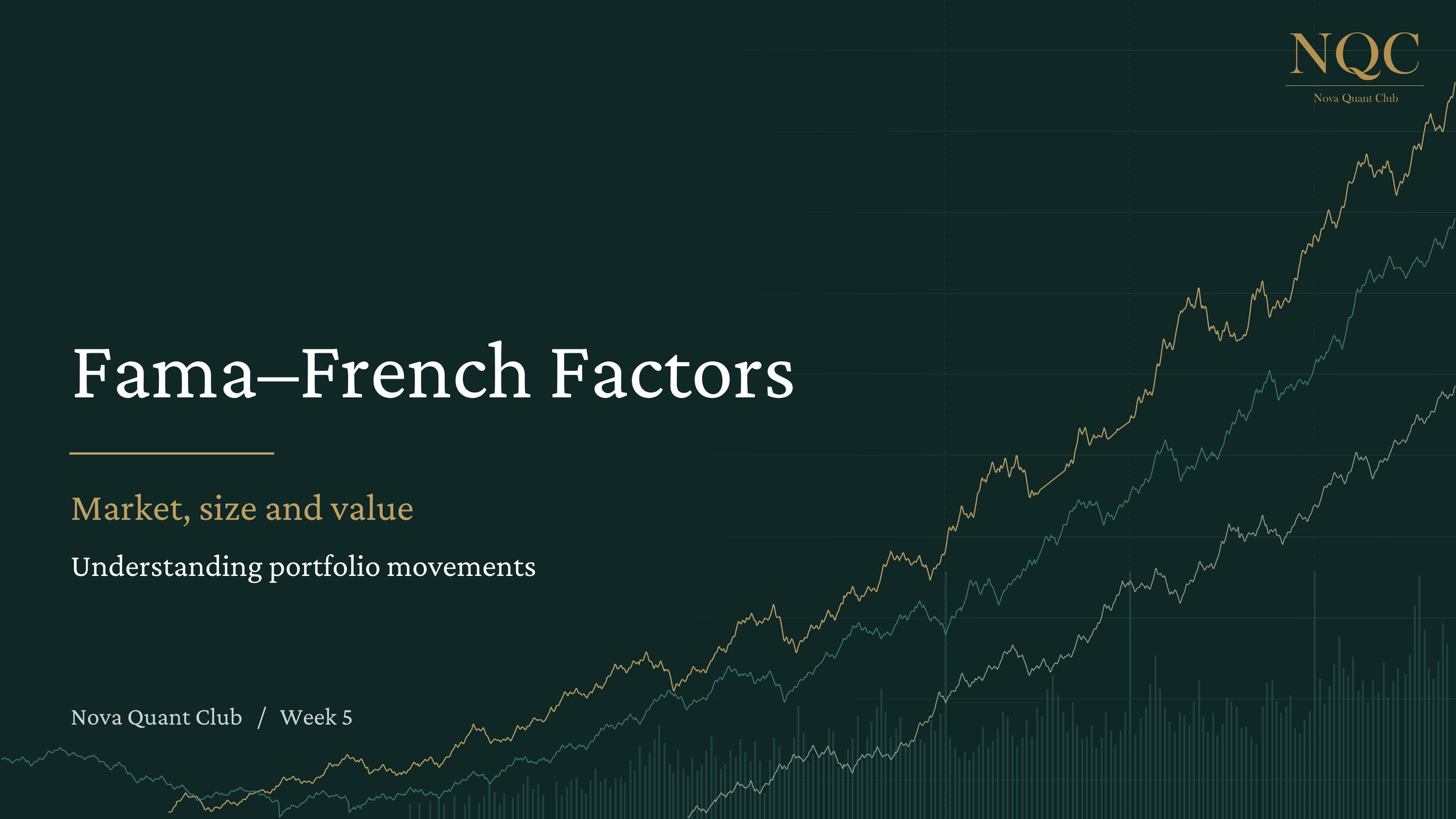
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Fama–French Factors

Market, size and value

Understanding portfolio movements

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CAPM recap

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The market as a common source of return movements

$$R_p - Rf = \alpha + \beta M\ (RM - Rf) + \varepsilon$$

Market excess return

The market's return above the risk-free return

Market beta

How strongly the portfolio moves with the market

Residual

The day's movement left outside the fitted return

CAPM’s limitations

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Market beta did not capture every average-return pattern

Market beta alone did not fully explain differences in average US stock returns

Size

Company size helped explain average-return differences

Book-to-market

Book-to-market helped explain average-return differences

Source: Fama & French (1992), The Journal of Finance
US stocks, 1963–1990

Fama and French’s approach

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Turn company characteristics into common return factors

$$R_p - Rf = \alpha + \beta M\ (RM - Rf) + \beta SMB\ SMB + \beta HML\ HML + \varepsilon$$

Market

Market index excess return
SPY as our proxy

SMB

Small-cap returns
minus big-cap returns

HML

High B/M – low B/M
(value – growth)

B/M = book-to-market

SMB: Small Minus Big

A return spread between smaller and larger companies

LONG smaller stocks

Positive weights
sum to +1

—

SHORT larger stocks

Negative weights
sum to –1

SMB: Weighting the two sides

Long smaller stocks, short larger stocks

```
for ticker in big_caps:
    weights[ticker] = -big_caps[ticker] / total_big_caps

for ticker in small_caps:
    weights[ticker] = small_caps[ticker] / total_small_caps
```

HML: High Minus Low

High book-to-market versus low book-to-market

LONG value

High book-to-market
Lower price vs book value

—

SHORT growth

Low book-to-market
Higher price vs book value

HML: From data to portfolio weights

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Get price-to-book from yfinance, then invert it

Book-to-market = 1 / price-to-book

```
stock = yf.Ticker(ticker)
mtb = stock.info.get("priceToBook")
market_cap = stock.fast_info["market_cap"]

if mtb is not None and mtb > 0 and market_cap is not None:
    btm = 1 / mtb

for ticker in value_stocks:
    weights[ticker] = value_caps[ticker] / total_value_cap

for ticker in growth_stocks:
    weights[ticker] = -growth_caps[ticker] / total_growth_cap
```

Preparing the factor data

Align the daily Market, SMB and HML return series

```
# Inside build_factor_data()
market = market.rename("Market")
smb = smb.rename("SMB")
hml = hml.rename("HML")

factor_data = pd.concat(
    [market, smb, hml],
    axis=1
)

factor_data = factor_data.dropna()
```

One row per date, with complete Market, SMB and HML returns

Computing portfolio returns

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Calculate the investment returns, then match every date

Portfolio return = sum of (weight × stock return)

```
investment_returns = compute_returns(portfolio_weights, portfolio)
investment_series = pd.Series(investment_returns, name="Portfolio")

regression_data = pd.concat(
    [investment_series, factor_data],
    axis=1
)

regression_data = regression_data.dropna().sort_index()
```

Teaching baseline: 20% in each stock · Daily risk-free return assumed zero

Estimating the regression

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Fit Market, SMB and HML, then read the factor exposures

```
# Inside run_regression()
factors = sm.add_constant(factors)
model = sm.OLS(returns, factors).fit()

three_factor_model = run_regression(
    regression_data["Portfolio"],
    regression_data[["Market", "SMB", "HML"]]
)

print(three_factor_model.params)
```

const: daily alpha
Market, SMB, HML: factor exposures

Factor exposures: our portfolio

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Using the three-factor results from our notebook

SMB
β = +0.009

A +1 pp move in SMB is associated with a +0.009 pp change in our portfolio's fitted daily return.

HML
β = −0.446

A +1 pp move in HML is associated with a −0.446 pp change in our portfolio's fitted daily return.

Market
β = +1.030

A +1 pp market move is associated with a +1.030 pp change in our portfolio's fitted daily return.

Other factors held constant · Historical regression estimates

Equal weights · 66 daily observations · RF = 0 · Coefficients rounded

Reading the signs: portfolio tilts

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The sign shows direction; the magnitude shows sensitivity

SMB	HML	Market
$\beta > 0$ Small-cap tilt	$\beta > 0$ Value tilt	$\beta > 0$ Moves with the market
$\beta < 0$ Large-cap tilt	$\beta < 0$ Growth tilt	$\beta < 0$ Moves against it

Our portfolio: almost no size tilt · growth tilt · market $\beta \approx 1.03$

Near-zero SMB/HML: little estimated style tilt. Exposures are not portfolio weights.

HML: what does −0.446 mean?

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Direction, sensitivity and explained variation

Sign

Negative HML exposure suggests growth-like (low B/M) behaviour.

Magnitude

About 0.446 pp in fitted return per 1 pp change in HML.

R^2

How much historical return variation the full model explains.

Growth beats value by 1 pp → HML = −1 pp

$(-0.446) \times (-1 \text{ pp}) = +0.446 \text{ pp}$

HML contribution to fitted return · Market and SMB held constant
−0.446 does not mean 44.6% of movements are explained.

Alpha and R^2

Unexplained average return and historical variation explained

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Alpha

Average daily return left unexplained by the chosen factor exposures

Positive alpha alone does not establish investment skill

$R^2 = 0.80$

80% of historical return variation explained in this hypothetical sample

Higher sample fit does not establish better forecasts

Adding factors cannot lower ordinary R^2 on identical observations